Digital Preservation of Cultural Heritage using XR Technologies

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Executive Summary

// Rezső innen The project "Digital Preservation of Cultural Heritage using XR Technologies" aims to revolutionize how we interact with and preserve history. By leveraging Extended Reality (XR)—encompassing Virtual, Augmented, and Mixed Reality—this project focuses on creating high-fidelity digital twins of endangered cultural heritage sites. The proposal addresses the urgent need for non-destructive preservation and accessible education. Key outcomes include a standardized framework for digital heritage reconstruction, a cloud-based XR platform for remote exploration, and specialized tools for historical researchers. This summary highlights our interdisciplinary approach combining computer science with digital humanities to ensure that cultural artifacts remain accessible for future generations despite physical decay or environmental threats.

Technical and Scientific Description of the Project

1 The Project Topic and its Practical Relevance

// Sári innen The project focuses on the digital preservation and accessibility of cultural heritage sites using XR technologies. By integrating 3D scanning, photogrammetry, and immersive VR/AR environments, it enables the creation of digital twins of endangered monuments.

Two major challenges are addressed:

- Physical degradation caused by mass tourism and environmental factors
- Limited accessibility to remote or fragile heritage sites

The project operates across Mediterranean, Central-Eastern European, and Atlantic regions, combining technological innovation with cultural heritage expertise.

2 Project Contribution Beyond the State of the Art

Recent advances in XR technologies and digital twin systems have opened new possibilities in cultural heritage preservation. However, existing solutions are often limited to isolated case studies or indoor environments.

Current limitations include:

- Lack of scalability
- Limited outdoor applicability
- Weak integration between scientific documentation and public accessibility

The project advances the state of the art by:

- Applying XR technologies in complex outdoor archaeological contexts
- Integrating large-scale, multi-region datasets
- Combining scientific documentation with immersive public access

3 Project Objectives and Outcomes

The main objectives of the project are:

- Develop high-quality 3D digital representations of cultural heritage sites
- Enable virtual access through VR/AR technologies
- Reduce physical tourism pressure on vulnerable sites
- Establish standardized methodologies for digital heritage documentation
- Foster interdisciplinary collaboration between technology and humanities
- Support open science through accessible datasets and publications

In detail, the project aims to save lesser-known or rapidly deteriorating monuments using 3D scanning and VR/AR technology, allowing them to be visited without the physical burden of mass tourism. Key targeted achievements include:

- **Technological Validation:** Testing XR platforms in complex, outdoor environments (e.g., ruins, built heritage) to increase Technology Readiness Levels (TRL) and demonstrate real-world impact.
- **Open Science and Visibility:** Generating high-quality datasets, 3D models, and VR assets as part of an open access ecosystem, maximizing visibility, citation, and future funding opportunities.
- **International Network Expansion:** Strengthening EU consortium networks by bringing together technology hubs, scientific publishers, and top-tier heritage protection partners across the Mediterranean, Central-Eastern, and Atlantic European regions.

4 Original and Innovative Contributions

The project introduces innovative contributions by combining XR technologies with scalable digital heritage workflows.

Key innovations include:

- Use of high-fidelity digital twins as reusable cultural assets
- Integration of XR environments with real-world heritage preservation strategies
- Standardized workflows for outdoor heritage digitization

Further breakthrough areas include:

- **New XR Application Domain:** Shifting advanced 4D reality capture (recording motion, gesture, sound, and physiological data) from industrial and healthcare sectors directly into cultural heritage. This represents a highly innovative use-case that aligns with EU "cultural heritage digitisation" priorities.

- **Validation in Complex Scenarios:** Pushing the technological boundaries by validating scanning technologies on outdoor, deteriorating ruins under real-world conditions, rather than controlled lab environments.
- **Open Access Revenue and Science Models:** Implementing an innovative ecosystem where digital heritage + XR research acts as fuel for an open science platform, increasing accessibility for developers, startups, and museums.

5 Inter-, Multi-, or Trans-disciplinary Characteristics

The project adopts an interdisciplinary approach combining computer science, archaeology, cultural heritage studies, and digital humanities. Technological development is closely integrated with heritage expertise, ensuring both scientific accuracy and user accessibility. This approach enables effective collaboration across domains and enhances the societal relevance of the project outcomes.

This interdisciplinary synergy is manifested through the following pillars:

- **Technology and Publishing:** Arts et Métiers (CAPLAB) provides cutting-edge XR tools, while Springer Nature ensures that the resulting digital heritage trend reaches the widest possible academic and public audience through open science and open access mechanisms.
- **Mediterranean Heritage Experts:** Sapienza Università di Roma and the National and Kapodistrian University of Athens bridge classical Greco-Roman archaeology, urban conservation, and climate risk assessments with modern digital documentation methods.
- **European Regional Synergy:** ELTE BTK (Hungary) brings AI-based processing and digital humanities expertise (Danube-AI), while Sorbonne Université (France) coordinates data collection on medieval fortifications and coastal erosion, ensuring a truly pan-European and cross-disciplinary framework.

Impact and Dissemination of the Project Results

1 Dissemination and Exploitation

Dissemination focuses on open science and wide accessibility:

- Open access publications (Springer Nature)
- Public datasets (3D models, scans)
- VR/AR applications for public engagement

Target audiences include:

- Researchers
- Cultural institutions
- Developers and industry stakeholders

2 Possible Applications with Market Potential

The market potential and exploitation strategies are deeply intertwined with the open science model and XR sector expansion:

- **Open Access Revenue Model:** The partnership with Springer Nature leverages Article Processing Charges (APCs) to ensure free access to publications, 3D models, and VR assets. This attracts not only academics but also software developers and XR startups, leading to higher download rates and increased brand reputation.
- **XR Technology Expansion:** Validating 4D capture in complex cultural heritage scenarios opens a completely new market segment for XR technologies, increasing their societal relevance and commercial viability beyond traditional industrial/health training.
- **Sustainable Tourism Economy:** By addressing the physical destruction caused by overtourism, the project provides a scalable digital alternative. "Experience-replicas" and remote access tools offer tourism boards new digital products that generate engagement while unburdening fragile UNESCO sites.

3 Improvements in Quality of Life

The project delivers impact on multiple levels:

Scientific:

- New methodologies in digital heritage
- Interdisciplinary research outputs

Technological:

- Validation of XR technologies
- Advancement of digital twin applications

Societal:

- Reduced tourism pressure
- Increased accessibility to heritage

European:

- Strengthened research networks
- Contribution to EU digital heritage strategies

4 Integration in Partner Strategies

Consortium Description

1 Project Director

2 Consortium Structure

The consortium consists of interdisciplinary partners:

Technology Partner: Arts et Métiers (CAPLAB / Laval Virtual) – XR technologies and 4D data capture

Publishing Partner: Springer Nature – Open access dissemination and scientific publishing

Mediterranean Region:

- Sapienza Università di Roma – archaeology and heritage theory
- National and Kapodistrian University of Athens – outdoor heritage and risk analysis

Central-Eastern Europe:

- Eötvös Loránd University (ELTE) – digital humanities and AI-based heritage processing

Atlantic Region:

- Sorbonne Université – data collection coordination and historical architecture

3 Research Team Leaders

4 Team Structure

5 Complementarities and Synergies

Project Management

1 Work Plan, Deliverables and Load Balancing

The project is structured into the following work packages:

A detailed timeline (Gantt chart) will define milestones and deliverables across the project duration.

2 WP1: Project Management and Coordination

3 WP2: Data collection and fieldwork

4 WP3: XR development and integration

5 WP4: XR Platform Development and UX Design

Leading Partner: Arts et Métiers (CAPLAB / Laval Virtual)

Objectives and Concept: This Work Package integrates advanced XR (VR/AR) and 3D technologies to create a new use-case for cultural heritage and tourism. By utilizing CAPLAB's 4D reality capture capabilities (motion, gesture, sound, physiological data), the project aims to preserve "artisanal intelligence" alongside physical structures. Moving the technology from industrial and health sectors into heritage increases its societal and market relevance, and allows for technological validation (increasing TRL) in complex, outdoor environments.

Deliverables / Schedule (from Gantt chart):

- **D4.1 Development of a VR-based "Virtual Tour" engine (Unreal Engine/Unity):** Year 1 (Month 10-12) & Year 2 (Month 1-3). This task involves building the core 3D rendering architecture capable of handling high-fidelity photogrammetry datasets smoothly. The engine will focus on optimizing lighting, texture loading, and spatial audio to maximize immersion without requiring industrial-grade hardware from the end users.

- **D4.2 User Experience (UX) research and prototype testing with focus groups:** Year 2 (Month 1-3). The team will conduct iterative usability testing with diverse demographic groups to identify pain points in virtual navigation. The goal is to ensure the interface is highly intuitive, accessible, and inclusive for both digital natives and older audiences.

- **D4.3 Development of an AR-based "On-Site Time Machine" module (ruin reconstruction via mobile application):** Year 2 (Month 4-6). This task focuses on augmenting the physical visitor experience by overlaying digital reconstructions onto real-world ruins in real-time. It requires precise spatial tracking and geolocation integration to seamlessly blend the 3D models with the user's camera feed.

6 WP5: Overtourism Management and Virtual Flow Control

Leading Partner: Sorbonne Université (with strong involvement from Sapienza Università di Roma and NKUA for Mediterranean use-cases)

Objectives and Concept: This Work Package directly addresses the pressing issue of mass tourism on vulnerable monuments in the Mediterranean and Atlantic regions. By establishing virtual alternatives, the physical burden on highly visited sites is reduced, and tourism flows can be diverted to lesser-known locations. This involves climate-sensitive digitization (e.g., coastal erosion monitoring) and utilizing the "Virtual-First" tourism model.

Deliverables / Schedule (from Gantt chart):

- **D5.1 Development of a "Virtual-First" tourism model (remote access vs. on-site restrictions):** Year 2 (Month 1-6). The aim here is to create strategic guidelines that prioritize digital exploration for highly vulnerable sites while limiting physical entry. Researchers will analyze data from Mediterranean use-cases to determine optimal carrying capacities and define conditions where virtual access becomes the primary mode of visitation.

- **D5.2 Design of digital ticketing system and "capacity monitoring" algorithms:** Year 2 (Month 4-6). This involves developing software tools that dynamically control physical ticket availability based on real-time sensor data and site deterioration levels. The algorithm will automatically incentivize virtual tickets or reroute visitors to nearby, less burdened attractions during peak hours.

- **D5.3 Awareness campaign about the "responsible digital tourist" concept:** Year 2 (Month 7-9). A comprehensive communication strategy will be rolled out to educate the public on the environmental and cultural importance of reducing physical tourism footprints. Through targeted media and social platforms, the campaign will promote VR/AR tours as premium, sustainable alternatives rather than mere substitutes.

7 WP6: Community Engagement and Social Impact

Leading Partner: Eötvös Loránd Tudományegyetem (ELTE BTK)

Objectives and Concept: WP6 ensures that the research outcomes and digital assets reach a broad audience, fostering strong societal connections. Driven by ELTE's digital humanities expertise, this WP involves local populations, integrates local legends into the XR narratives, and promotes an open science ecosystem where datasets and 3D models become accessible. The ultimate goal is to embed the digital heritage into community identities and measure the resulting social impact.

Deliverables / Schedule (from Gantt chart):

- **D6.1 Community workshops on the relationship between local identity and digital representation:** Year 1 (Month 4-12). These interactive sessions will actively engage residents, local historians, and policymakers to discuss how their regional heritage is being digitized. The objective is to ensure that the 3D assets and virtual narratives accurately and respectfully reflect the community's own cultural self-understanding.
- **D6.2 Co-creation processes: integrating local legends into the XR narrative:** Year 1 (Month 7-12). By utilizing crowdsourced oral histories and regional folklore, the team will enrich the digital reconstructions with authentic, community-driven storytelling. This transforms the VR experience from a sterile architectural walkthrough into a deeply engaging, living historical narrative.
- **D6.3 Preparation of a social impact assessment (SIA):** Year 2 (Month 10-12). This final deliverable will systematically measure and document how the project's digital interventions have influenced local economies, community cohesion, and public attitudes toward heritage conservation. The resulting report will provide empirical data to validate the societal relevance of the "Digital Heritage + XR" approach.

8 WP7: Economic Models and Revenue Generation

Leading Partner: Springer Nature

Objectives: The primary objective of this Work Package is to establish a robust financial framework that ensures the project's self-sufficiency beyond the initial 24-month funding period. We aim to transform high-fidelity digital twins into viable economic assets for local communities and the consortium by creating diverse revenue streams.

Duration: Month 7-12

Tasks:

- **Task 7.1: Digital ticketing and subscription models (B2C):**

  Development of a tiered access system for the VR platform. This includes "Basic" (free educational access), "Premium" (high-resolution guided tours), and "Seasonal Pass" (access to all regional clusters).
- **Task 7.2: Development of a licensing framework for educational institutions and museums (B2B):**

  Designing a standardized licensing package for third-party stakeholders such as schools, universities, and international museums to use the VEP engine for their own curricula or exhibitions.
- **Task 7.3: Integrating local artisanal products and services into the virtual marketplace:**

  Creating a digital bridge between virtual visitors and local craftspeople. This involves an e-commerce layer within the XR experience allowing users to purchase physical goods from the CEE, Mediterranean, or Atlantic regions.

Deliverables:

- **Deliverable 7.1: Comprehensive Business Sustainability Plan:**

  5-year financial projection including revenue streams and operational costs.

- **Deliverable 7.2: Standardized Licensing Agreement Template:**

  Legal documentation for B2B partnerships.

Milestones:

- **Milestone 7:**

  Execution of the first commercial license agreement with a non-consortium educational entity (Month 20).

Geographical Relevance: Revenue models will be tailored to regional strengths: emphasizing the "Digital Souvenir" market in the Mediterranean and "Cultural Educational Kits" in the CEE region.

9 WP8: Testing, Validation and Pilot Actions

Leading Partner: Arts et Métiers (CAPLAB / Laval Virtual)

Objectives: This phase focuses on the technical and social validation of the developed prototypes. By deploying the platform in diverse geographic settings, we ensure the technology is resilient to varying environmental factors and user backgrounds, while measuring the actual impact on physical site preservation.

Duration: Month 4-12

Tasks:

- **Task 8.1: Implementation of Multi-Regional Pilot Events:**

  Organizing four physical and virtual "Launch Days" across the Atlantic (focusing on coastal erosion sites), Mediterranean (ancient ruins), and CEE (castles and wooden churches).

- **Task 8.2: Technical Performance and Stress Testing:**

  Evaluating rendering performance across different devices (from high-end VR headsets to mobile AR) and network bandwidth conditions.

- **Task 8.3: Comprehensive User Feedback Collection:**

  Utilizing User Experience (UX) research methods, including eye-tracking during VR sessions and qualitative focus groups with diverse age demographics to refine the interface.

Deliverables:

- **Deliverable 8.1: Pilot Evaluation Report:**

  A technical summary of the performance metrics and bug reports gathered during the pilots.

- **Deliverable 8.2: Social Impact and UX Assessment:**

  Detailed analysis of how the virtual experience shifted visitor attitudes toward physical site conservation.

Milestones:

- **Milestone 8:**

  Successful completion of all four regional pilot actions with a minimum 85% positive user satisfaction rate (Month 22).

Geographical Relevance: Testing will address specific regional challenges, such as high-glare lighting conditions for AR in Greece and hardware resilience in high-humidity Atlantic environments.

10 WP9: Sustainability, Dissemination and Communication

Leading Partner: Springer Nature

Objectives: Following the **CHARM-EU** model of long-term alliance strategy, this WP focuses on the post-project life cycle. It ensures the alliance remains sustainable through the repurposing of existing resources, the establishment of a permanent legal identity, and the wide dissemination of scientific results.

Duration: Month 1-12

Tasks:

- **Task 9.1: Scientific Dissemination and Open Science:**

  Authoring peer-reviewed publications and presenting at international heritage and technology conferences (e.g., EuroVR, Digital Heritage).

- **Task 9.2: Closing Conference and "Virtual-Europe" Exhibition:**

  A flagship event in Brussels to showcase the project results to EU policymakers, demonstrating how XR can mitigate overtourism.

- **Task 9.3: Exploitation Plan and Long-Term Alliance Strategy:**

  Developing the "Long-term strategy for sustainability of the alliance." This includes exploring the creation of a separate legal entity (e.g., an AISBL) to manage IP and maintain the digital repository beyond the pilot phase.

Deliverables:

- **Deliverable 9.1: Final Sustainability and Exploitation Plan:**

  A detailed roadmap for the alliance's activities for the next 10 years, including financial commitments from each member.

- **Deliverable 9.2: Communication Toolkit:**

  A package of branding materials and media assets for regional partners to ensure consistent messaging.

Milestones:

- **Milestone 9:**

  Final Dissemination Event and publication of the VEP White Paper on Digital Heritage Policy (Month 24).

Sustainability Strategy (Alliance Specifics):
In line with the strategy of the alliance, members are committed to action beyond the project lifecycle. This involves:

- **Repurposing existing consortium resources** to ensure continuity without requiring significant new budget requirements.

- **Investigating legal formulas** (such as a separate legal identity) to overcome bureaucratic or management barriers to long-term cooperation.

- **Leveraging existing national and European grants** to support ongoing student and researcher mobility between the Mediterranean, Atlantic, and CEE partner regions.

11 Coordination and Task Schedule

The project timeline spans 24 months and is structured through a phased implementation of Work Packages. A detailed Gantt chart defines task dependencies and scheduling across all WPs.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP 1: Management of the project	1.1	Administrative and financial management	Sapienza Università di Roma								
	1.2	Development of a Quality Assurance and Risk Management Plan									
	1.3	Data Management Plan (DMP) based on GDPR and Open Science principles									
WP2: Methodological Framework and Site Selection	2.1	Development of a vulnerability and tourism impact index.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	2.2	Multi-stakeholder analysis (involving local population, local governments, academia)									
	2.3	Finalization of site selection matrix (aspects: degree of deterioration, limited accessibility).									
WP3: High-Fidelity Data Acquisition and 3D Scanning	3.1	Laser scanning (LiDAR) and aerial photogrammetry (drone-based data collection).	Nemzeti Kapodisztriasz Athéni Egyetem (NKUA)								
	3.2	Texture optimization and material-specific digital preservation.									
	3.3	Metadata enrichment (integration of archaeological and art historical data).									
WP4: XR Platform Development and UX Design	4.1	Development of a VR-based "Virtual Tour" engine (Unreal Engine/Unity).	Arts et Métiers (CAPLAB / Laval Virtual)								
	4.2	User Experience (UX) research and prototype testing with focus groups.									
	4.3	Development of an AR-based "On-Site Time Machine" module (ruin reconstruction via mobile application).									

Figure 1: Project schedule (Gantt chart) of WP1, WP2, WP3 and WP4 for the 24-month period.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP5: Overtourism Management and Virtual Flow Control	5.1	Development of a "Virtual-First" tourism model (remote access vs. on-site restrictions).	Sorbonne Université								
	5.2	Design of digital ticketing system and "capacity monitoring" algorithms.									
	5.3	Awareness campaign about the "responsible digital tourist" concept.									
WP6: Community Engagement and Social Impact	6.1	Community workshops on the relationship between local identity and digital representation.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	6.2	Co-creation processes: integrating local legends into the XR narrative.									
	6.3	Preparation of a social impact assessment (SIA).									
WP7: Economic Models and Revenue Generation	7.1	Digital ticketing and subscription models (B2C).	Springer Nature								
	7.2	Development of a licensing framework for educational institutions and museums (B2B).									
	7.3	Integrating local artisanal products and services into the virtual marketplace.									
WP8: Testing, Validation and Pilot Actions	8.1	Organizing pilot events in 4 countries (on-site and remote testing).	Arts et Métiers (CAPLAB / Laval Virtual)								
	8.2	Technical performance evaluation (loading speed, rendering accuracy).									
	8.3	Collect user feedback using quantitative and qualitative methods.									
WP9: Sustainability, Dissemination and Communication	9.1	Scientific publications and conference participation (Open Access).	Springer Nature								
	9.2	Organization of a closing conference and "Virtual Heritage" exhibition.									
	9.3	Development of an Exploitation Plan to adapt the technology to other regions.									

Figure 2: Project schedule (Gantt chart) of WP5, WP6, WP7, WP8 and WP9 for the 24-month period.

12 Research Infrastructure

The project relies on a comprehensive network of specialized infrastructures provided by the consortium:

- **CAPLAB Platform (Arts et Métiers / Laval Virtual):** A European hub for XR, digital twins, and 3D technologies, providing robust infrastructure for 4D reality capture and technology transfer.
- **Digital Open Science Ecosystem (Springer Nature):** The platform infrastructure for hosting and distributing open access articles, datasets, 3D models, and VR assets.

- **Mediterranean Field Sites (Sapienza & NKUA):** Access to highly vulnerable urban UNESCO sites in Rome and weather-exposed open-air ruins in Greece, serving as the physical infrastructure for technological testing and validation.
- **Digital Humanities Network (ELTE BTK):** The National Laboratory for Digital Heritage and the Danube-AI initiative in Hungary, supplying the computational and methodological infrastructure for AI-driven heritage processing.
- **Data Collection Network (Sorbonne Université):** Established infrastructure for coordinating architectural and historical documentation, particularly concerning fortifications and coastal heritage.

13 Manpower Allocation

The consortium partners provide highly specialized manpower and expertise:

- **Arts et Métiers (CAPLAB / Laval Virtual):** Acts as the primary technology partner, offering engineering expertise in XR, VR/AR, and digital twins for industrial cooperation and technology transfer.
- **Springer Nature:** One of the world's largest scientific publishers, dedicating personnel to manage open access dissemination, data sharing, and maximizing the project's global impact.
- **Sapienza Università di Roma:** Provides researchers specialized in classical Greco-Roman archaeology, urban heritage conservation, and interdisciplinary monument protection theory.
- **National and Kapodistrian University of Athens (NKUA):** Contributes experts in classical and Byzantine archaeology, long-term site monitoring, and climatic risk analysis for open-air ruins.
- **Eötvös Loránd Tudományegyetem (ELTE BTK):** Supplies specialists in digital archaeology, digital humanities, and AI-based heritage processing from the National Laboratory for Digital Heritage.
- **Sorbonne Université:** Brings leading scholars from the Institut d'art et d'archéologie, focusing on medieval and early modern architectural history and data collection.
- **University of Bologna (Unibo) [Optional]:** Offers top-tier European expertise in medieval studies and heritage digitization via the FrameLab project.

14 Budget and Partner Shares

The budget is allocated across major categories:

- Personnel costs
- Equipment and technology (XR, scanning tools)
- Travel and fieldwork
- Dissemination and publication

Detailed financial planning ensures efficient resource allocation and alignment with project objectives.

References